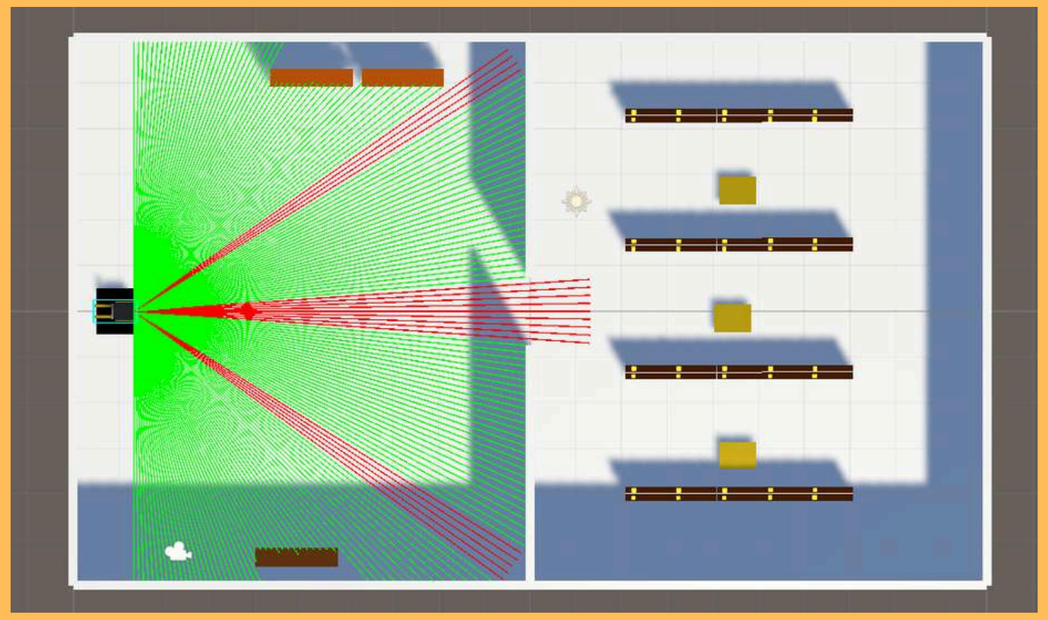
VIETNAMESE GERMAN UNIVERSITY

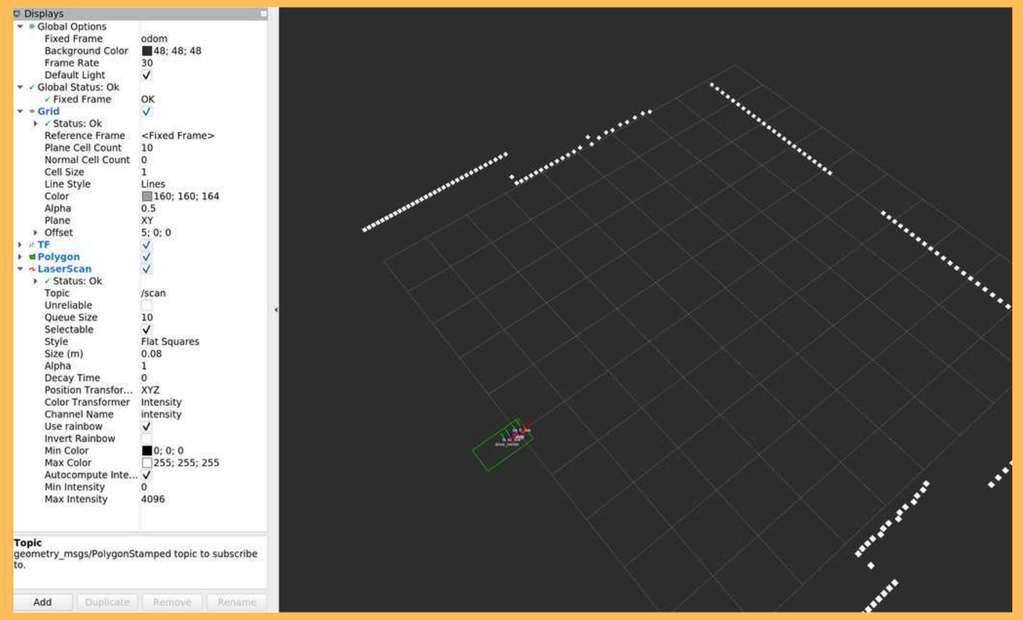
COLLABORATIVE AMR FOR WAREHOUSE-TO-AISLE STOCK REPLENISHMENT

WEEK 7 REPORT - GROUP 7

LAST WEEK'S RECAP



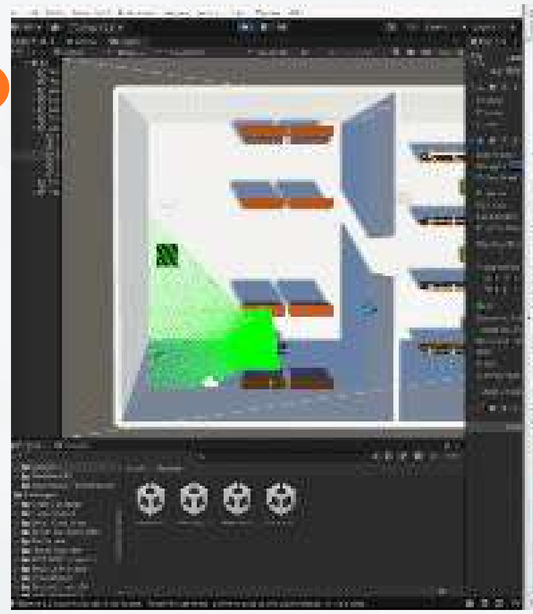
Set up Lidar on the AMR

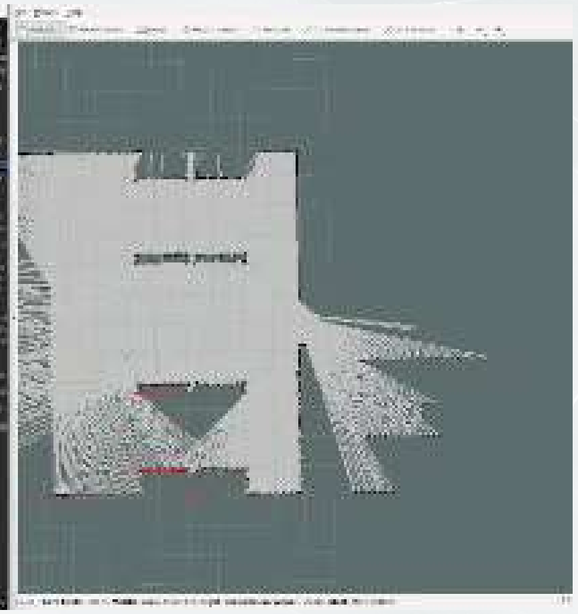


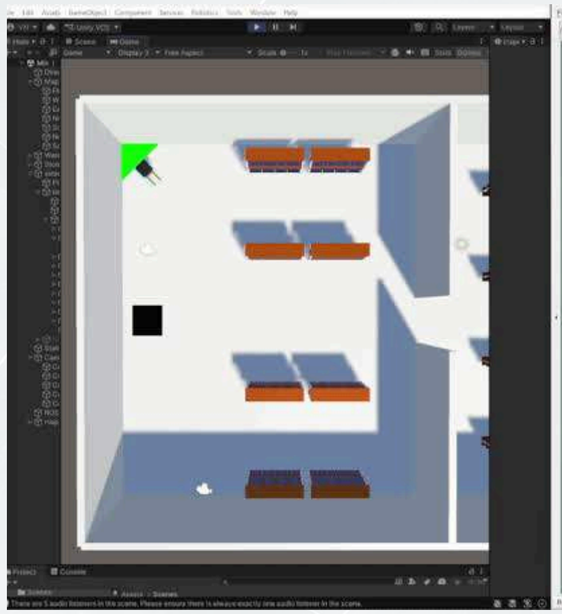
Publish Laser to /Scan

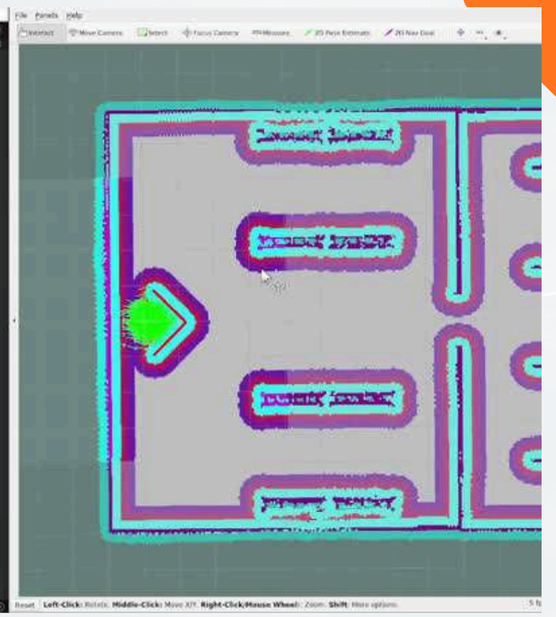
Problem: We need Spacial Awareness

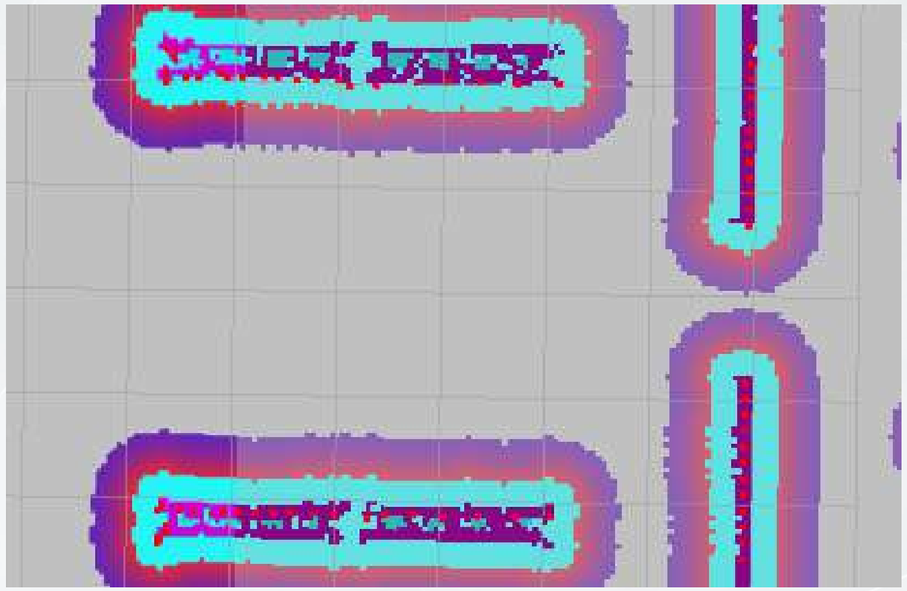
PROBLEMS THAT WE SOLVED THIS WEEK







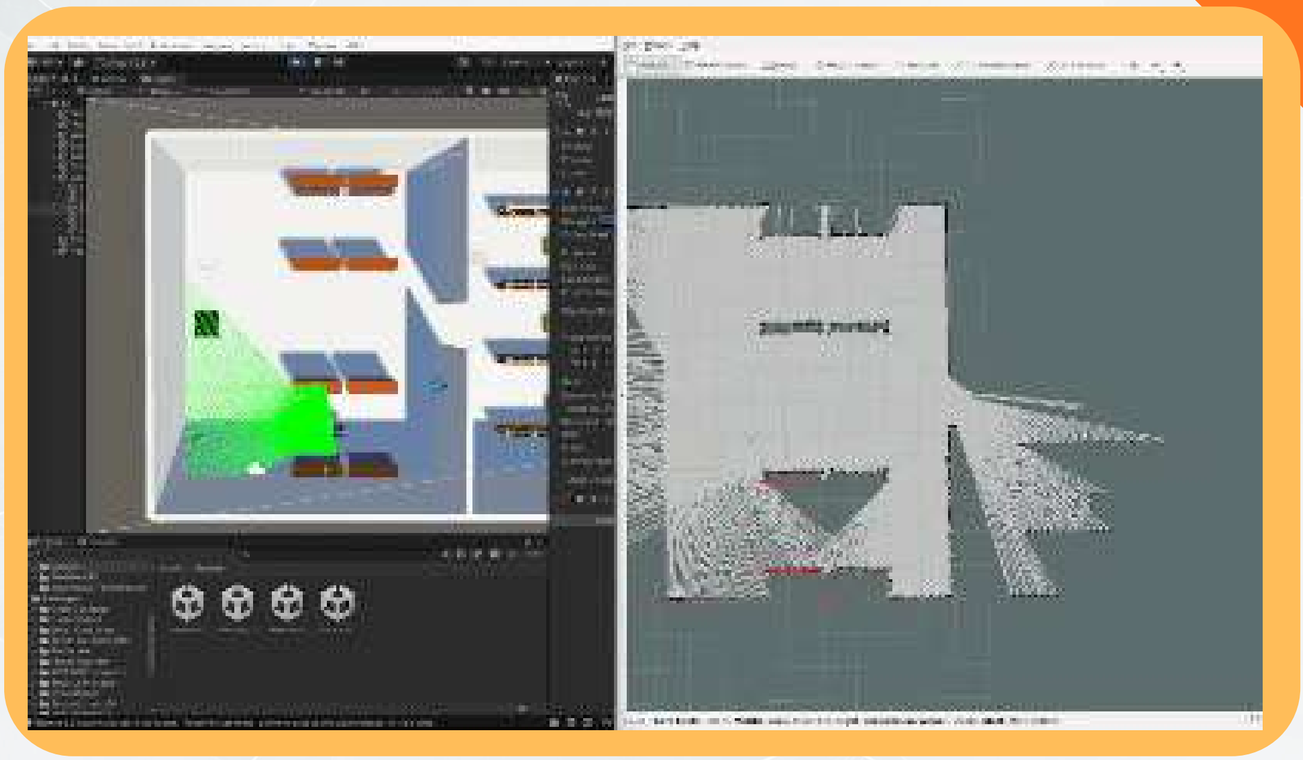




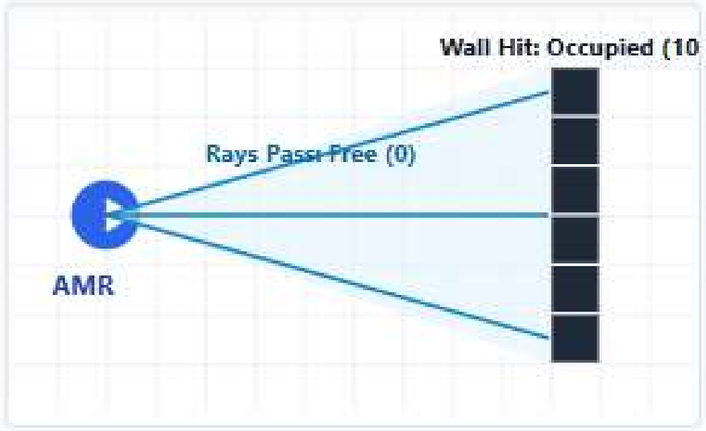
- Create a 2D Map of the Warehouse (SLAM / gmapping)
- Generate Safety Buffers via Costmap Inflation
- AMR Localization using AMCL (Particle Filter)

MAPPING VIA SLAM AND GMAPPING

- SLAM - Simultaneous Localization and Mapping
  - Create a map by tracking surrounding Landmarks and estimating movement via sensors (Lidar + Odometer).
  - → Gmapping



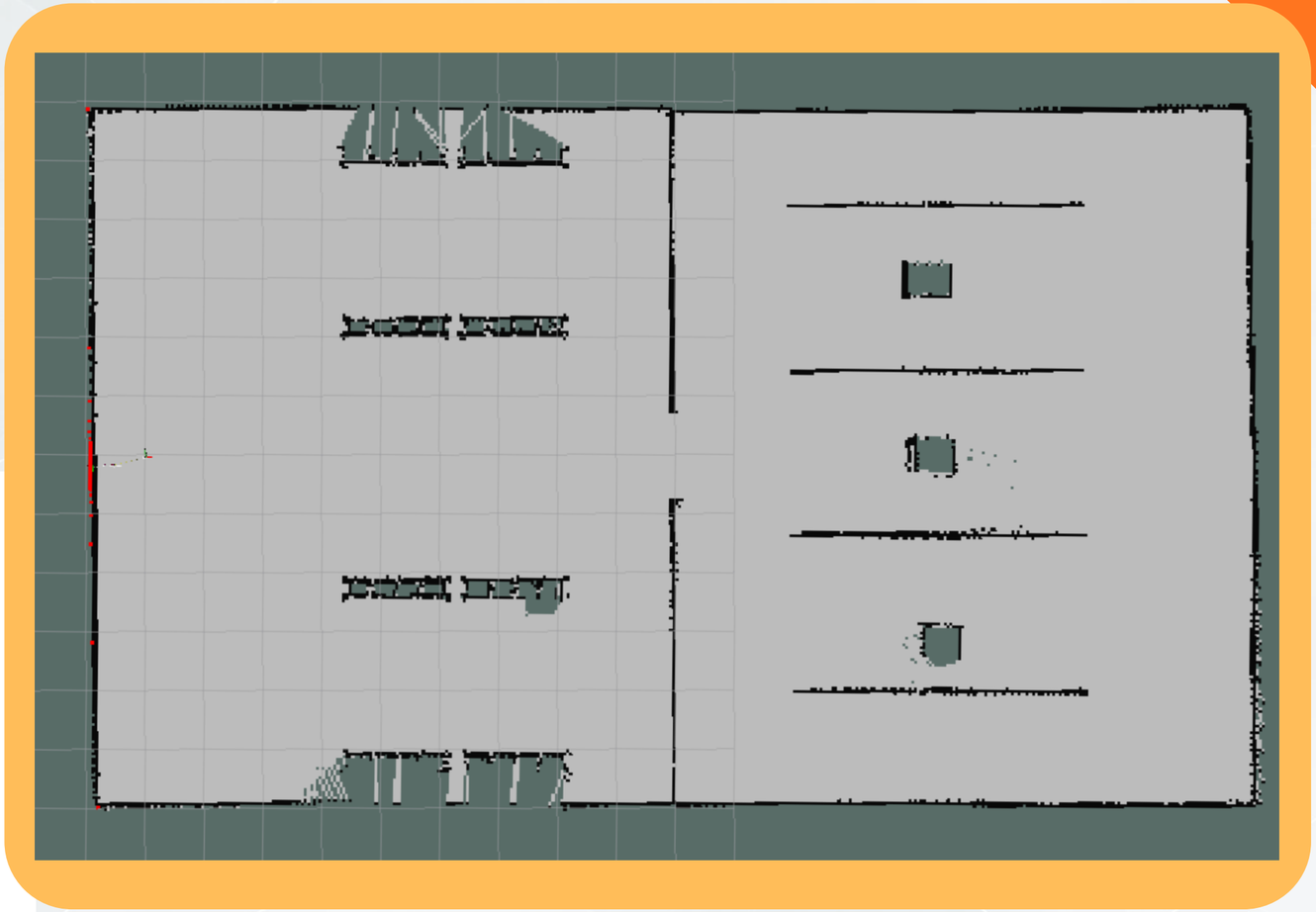
MAPPING VIA SLAM AND GMAPPING



1. Carving Free Space (Ray Tracing):
Every 5cm grid cell the laser travels through without hitting an obstacle is confirmed as empty floor (Cost = 0, White).

2. Marking Walls (Hit Point):
The exact cell where the ray reflects back is marked as a solid obstacle (Cost = 100, Black).

3. Log-Odds Reinforcement:
Multiple laser scans over time confirm permanent obstacles, filtering out dynamic noise and transient reflections.

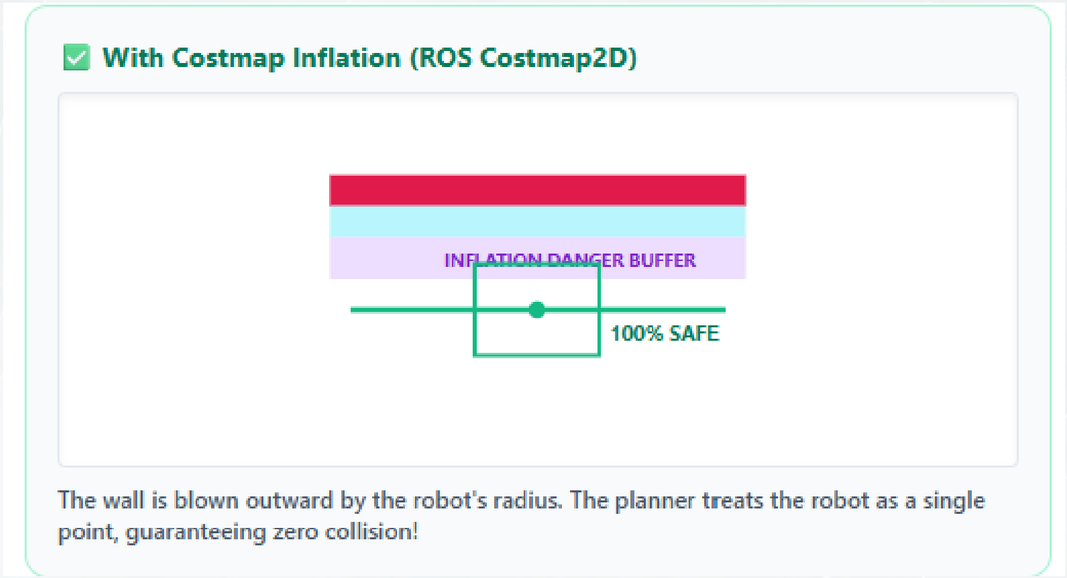


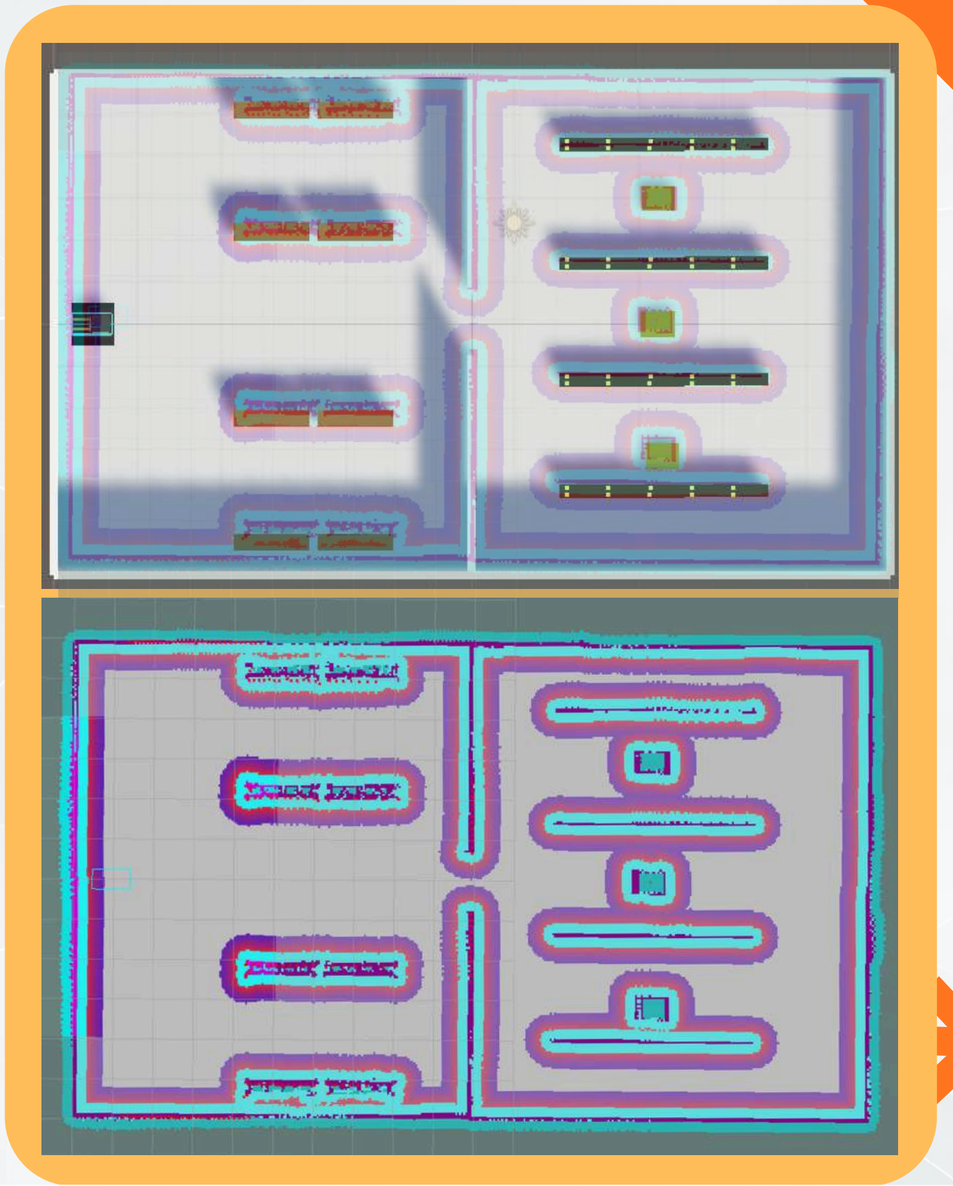
2D Map with gmapping

COSTMAP INFLATION FOR SAFETY BUFFER

- Path planners treat robots as zero-sized points
- But AMR are sized rectangles → risk scraping wheels and chassis against walls.

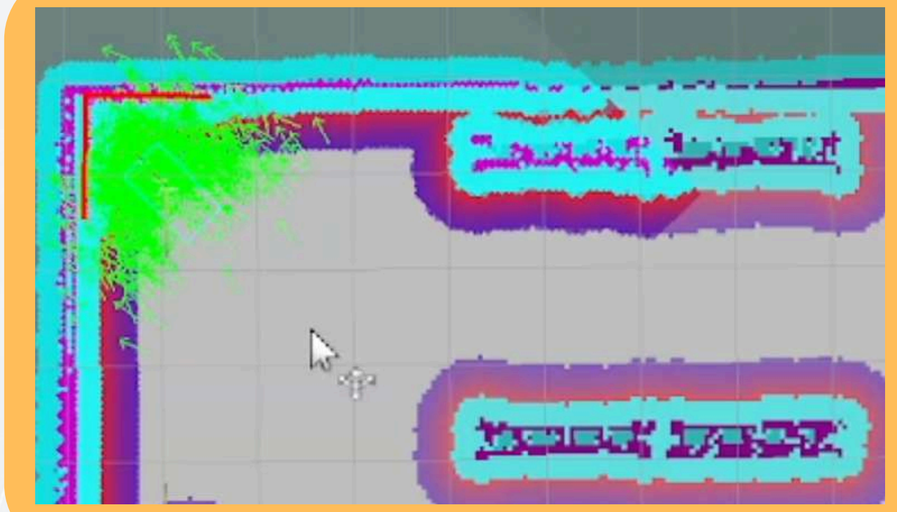
→ *Expanding obstacle boundaries by the robot’s physical radius.*

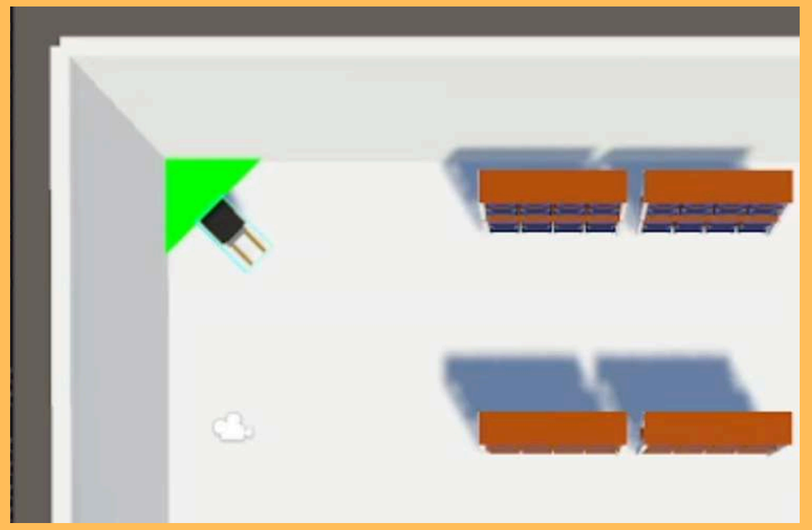




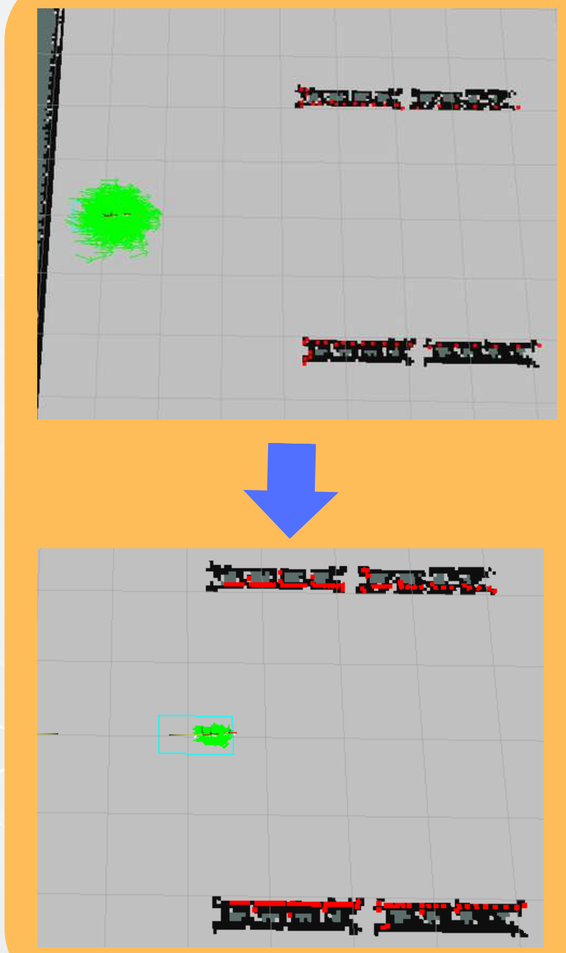
LOCALIZATION WITH AMCL

- AMCL (Adaptive Monte Carlo Localization).
- Lighter version of SLAM.
- Guess the AMR’s pose via saved Map





Big particle cloud because ambiguity



Particles collapsing

LOCALIZATION WITH AMCL

DIAGRAM 2: How a Particle Scores Its Guess Against the Pre-Saved Map

Why AMCL is fast: the map is frozen, so it only calculates ray distances.

Particle A: High Probability Candidate — Weight: 0.96

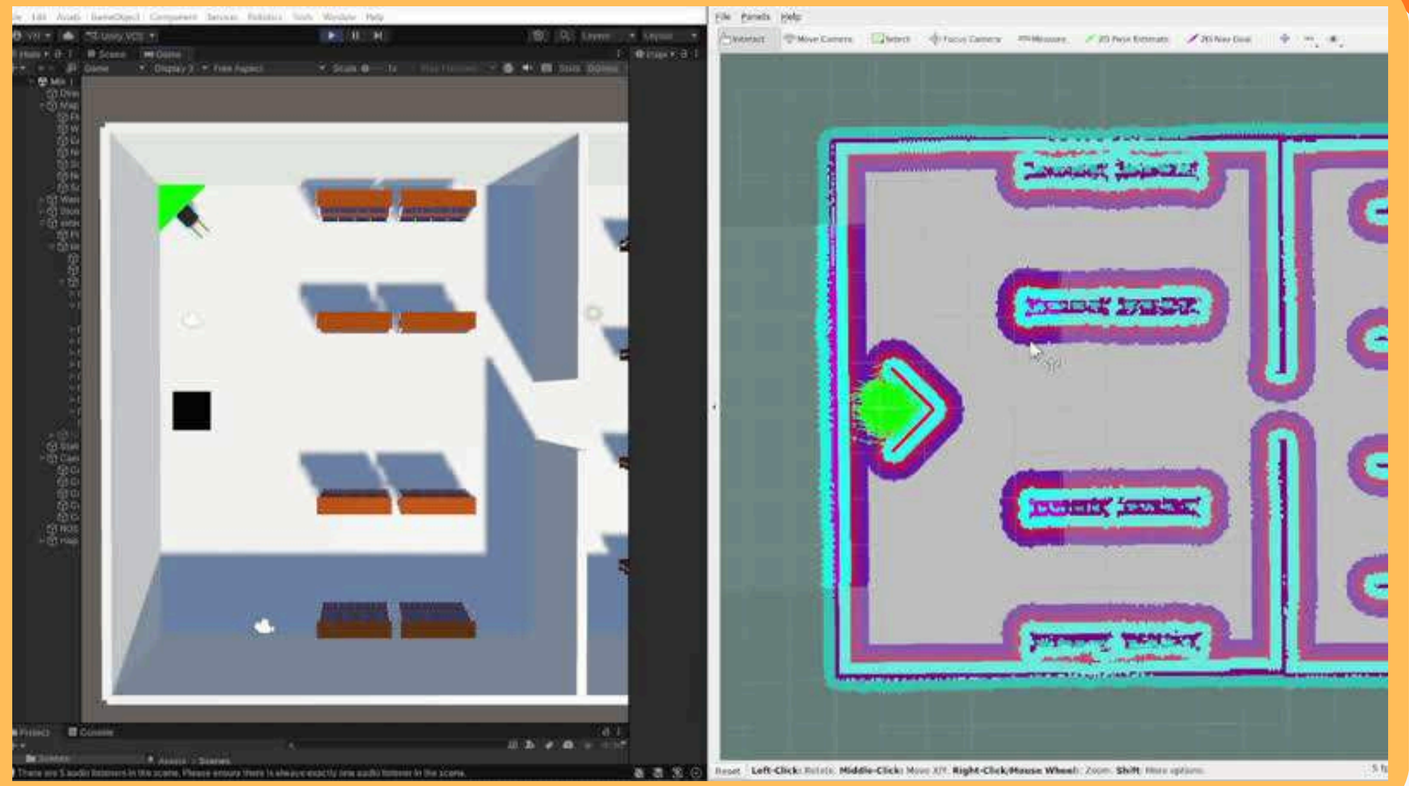


Prediction matches reality: Particle says wall should be at 2.5m, real LiDAR reports 2.5m. **Particle is cloned!**

Particle B: Incorrect Guess — Weight: 0.02



Mismatch: Particle expects empty space where a real wall exists. **Particle is deleted!**



PLANNING FOR NEXT WEEK

- Deploy move_base Navigation Stack
  - Integrate navfn (Global Dijkstra Planner).
- Configure and Tune dwa_local_planner (Dynamic Window Approach)
- Autonomous Point-to-Point Navigation Testing
  - Send 2D Nav Goals in RViz to verify autonomous driving and static obstacle avoidance in narrow warehouse aisles.

THANK YOU